



**Instituto Tecnológico de
Saltillo**



Ingeniería en Sistemas Computacionales

Computo en la Nube

7 am

**Actividad 4.1 – Desarrollar la práctica de escalada
de privilegios en Linux usando sudo.**

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```
root@srv:~# useradd usuario_test
root@srv:~# passwd usuario_test
New password:
Retype new password:
passwd: password updated successfully
```

Editado con visudo

```
# While you shouldn't normally run git as root, you need to with
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d

usuario_test ALL=(ALL) NOPASSWD:/usr/bin/nano

root@srv:~#
```

```
GNU nano 8.4 /etc/passwd *
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin
sshd:x:989:65534:sshd user:/run/sshd:/usr/sbin/nologin
debian:x:1000:1000:debian,,,:/home/debian:/bin/bash
mysql:x:101:105:MySQL Server:/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
polkitd:x:988:988:User for polkitd:/:/usr/sbin/nologin
usuario_test:x:1002:1002:/:/home/usuario_test:/bin/bash

[ Recording a macro... ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```

usuario_test@srv:~$ cat /etc/passwd
root::0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin
sshd:x:989:65534:sshd user:/run/sshd:/usr/sbin/nologin
debian:x:1000:1000:debian,,,:/home/debian:/bin/bash
mysql:x:101:105:MariaDB Server:/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
polkitd:x:988:988:User for polkitd:/:/usr/sbin/nologin
usuario_test:x:1002:1002:~/home/usuario_test:/bin/bash
usuario_test@srv:~$

```

```

-bash: visudo: command not found
usuario_test@srv:~$ su -
root@srv:~#

```

```

#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

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# User privilege specification
root    ALL=(ALL:ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@include /etc/sudoers.d

```

```

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-L
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E

```

```
root@srv:~#  
logout  
usuario_test@srv:~$ sudo nano /etc/passwd  
[sudo] password for usuario_test: _
```

```
GNU nano 8.4  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin
```

```
usuario_test@srv:~$ su -  
Password:
```

Conclusion

Es importante que como administrador de sistema no se den permisos innecesarios, esto para que los usuarios solo puedan realizar lo que tienen que realizar, esto se conoce como política de permisos mínimos, para que en caso de que un atacante consiga credenciales su impacto sea menor en el sistema.

Nunca se deben de dar permisos sin contraseñas en ningún momento, ya que la contraseña es la última protección en caso de que la cuenta sea comprometida, y si es posible se debe de configurar algún tipo de mfa.